



PhonePe Digital Payment Analytics

Power BI Dashboard Project — Aryan Rai, Data Analyst

An end-to-end business intelligence solution analyzing ₹3,333M in **transactions** across PhonePe's four core service categories – built to surface actionable insights for product, operations, and growth teams.

POWER BI

DAX

POWER QUERY

EXCEL

DATA ANALYTICS

Project Overview

Objective & Business Problem

Build an interactive Power BI dashboard to analyze PhonePe transactions and generate actionable business insights across four service verticals. The core challenge: understand transaction trends, monitor KPIs, identify payment failure patterns, and evaluate service performance – all from a single unified view.

Dataset

~50,000 transaction records in Excel covering Insurance, Loans, Money Transfer, and Recharge & Bills – January through December 2024.

Deliverables

5-page interactive dashboard (Home, Insurance, Loans, Money Transfer, Recharge & Bills) with KPI cards, slicers, bar charts, pie charts, and area charts.

Tools & Stack

Excel

Raw data source & cleaning

Power BI

Dashboard & visualization

Power Query

ETL & transformation

DAX

Calculated measures & KPIs

Dataset Architecture & Data Model

Table Structure

1
All_Users Master user table – User_ID, Name, Age, Join_Date. Acts as primary lookup for relationship modeling.
2
All_Transactions Unified transaction log – Transaction_ID, Amount, User_ID, Service, Service Type, Payment Status, Reason, Date.
3
Service Tables Recharge_Bills, Money_Transfer, Loans, Insurance – each with category-level columns (Loan Type, Transfer Type, Insurance Type, etc.).

Data Model & Cleaning

The model follows a **star schema** – All_Users and All_Transactions serve as fact/dimension tables linked to service-specific tables, enabling robust cross-filter analysis in Power BI.

→ **Removed duplicates**

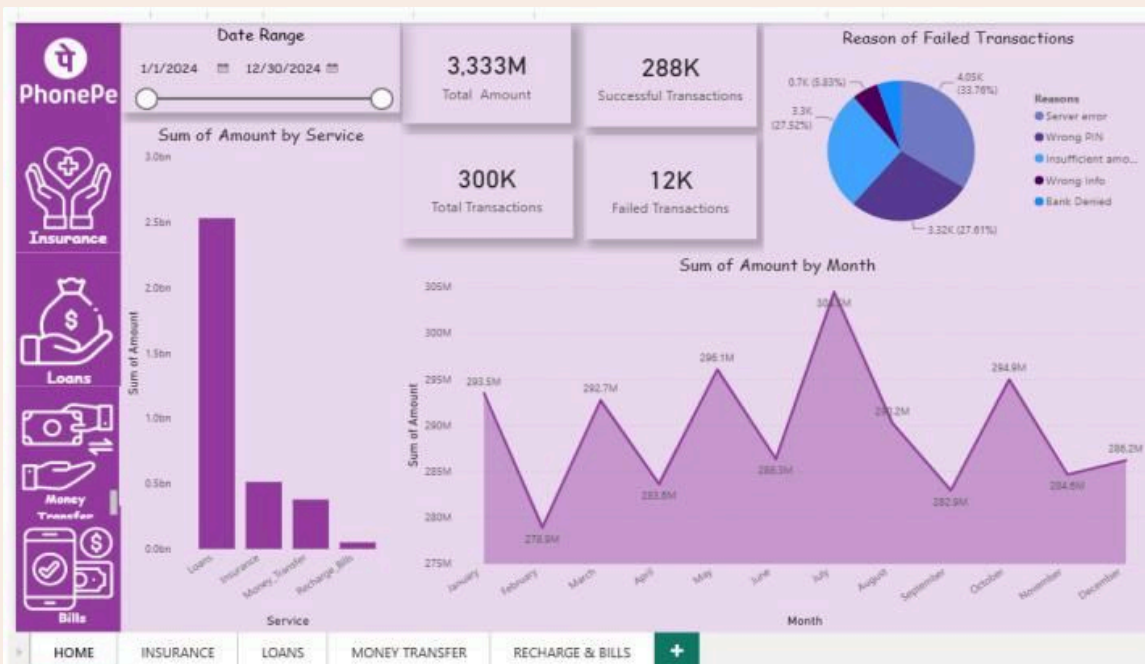
Ensured transaction-level uniqueness across all 50K records.

→ **Type corrections & missing values**

Corrected data types, handled nulls, transformed date fields.

→ **Month column via Power Query**

Derived for time-intelligence DAX measures and monthly trend visuals.



Home Dashboard — Transaction Overview

₹3,333M

Total Transaction Amount

Across all services, Jan-Dec 2024

96.75%

Success Rate

Platform-wide payment reliability

Top Failure Reasons

300K

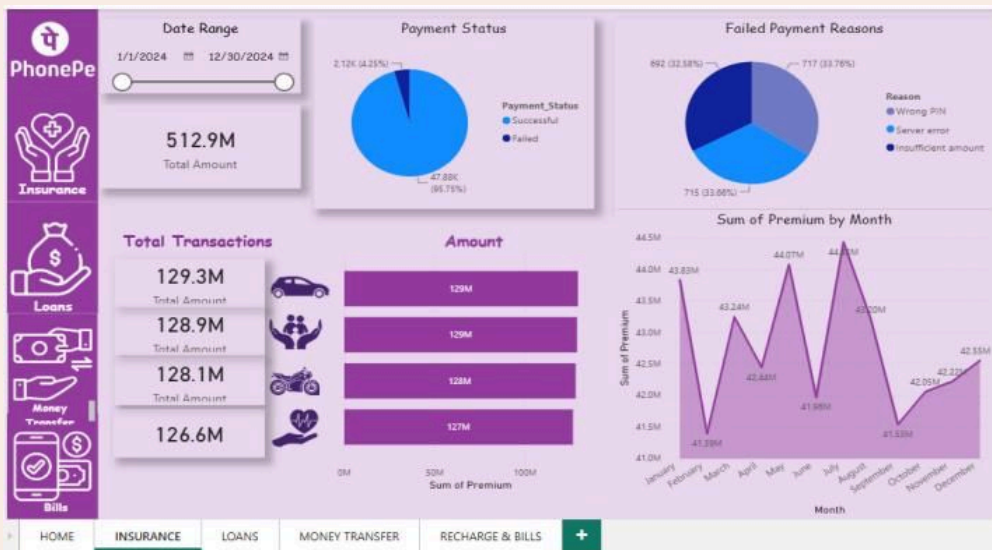
Total Transactions

288K successful · 12K failed

Jul

Peak Month

₹333.3M – Feb was lowest at ₹278.9M



Insurance Dashboard

₹512.9M

Total Premium Collected

Jan-Dec 2024, all insurance types

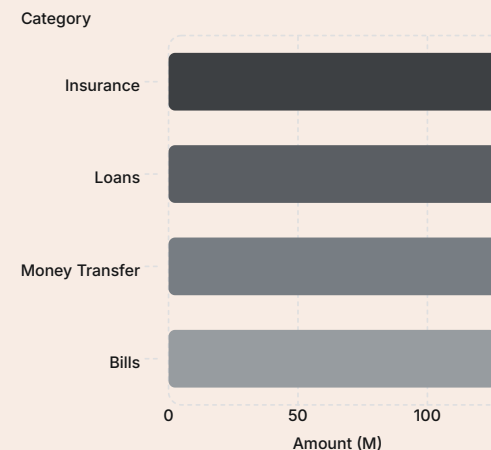
96.75%

Success Rate

47.80K successful · 2.12K failed transactions

All four service categories contribute near-equally to total platform volume, underscoring PhonePe's balanced multi-vertical revenue model. Insurance leads marginally at ₹129.3M.

Service Category Contribution (₹M)





Loans Dashboard

Loan Types Tracked

Gold Loan Highest individual transaction amounts	Mutual Fund High-value transactions alongside Gold
Auto Loan Unique disbursement patterns	Credit Score Usage-linked loan eligibility tracking

DAX Measures Applied

- Total Loan Amount
- Success Rate %
- Monthly Loan Trend
- Failure Count by Reason

Key Findings

~₹1.03M/month baseline Consistent loan disbursement throughout the year – minimal seasonal variance except November.
₹2.03M spike — November 2024 Likely driven by festive season lending demand; 2× the monthly baseline, warrants infrastructure pre-planning.
Primary failure: Payment Info errors Data entry or validation failures at submission – a UX/form-validation improvement opportunity.



Money Transfer Dashboard

₹378.2M

Total Transfer Volume

All methods combined, Jan-Dec 2024

100%

Failure Cause

Insufficient Amount accounts for all failed transfers

~₹30M

Per-Method Contribution

UPI ID · Bank Account · QR Code · Mobile Number – near-equal split

₹2.03M

November Peak

Consistent with platform-wide seasonality; baseline ~₹1.03M/month

Transfer Methods



To QR Code

~₹30M – contactless merchant payments



To Bank Account

~₹30M – IMPS/NEFT equivalent transfers



To UPI ID

~₹30M – P2P UPI address transfers



To Mobile Number

~₹30M – phone-linked instant transfers

Visualization Approach

→ Line chart for monthly trends

Visualizes the baseline-to-November spike pattern across the full year.

→ Bar chart for method comparison

Highlights the equal contribution across all four transfer channels.

→ Slicers for date range filtering

Enables drill-down by month, quarter, or custom range for ad hoc analysis.



Recharge & Bills Dashboard

₹48.7M

Total Processed

All bill categories, Jan-Dec 2024

~₹10M

Per-Category Split

Electricity · DTH · Mobile · Cable TV – near-equal distribution

Key Business Insight

Uniform distribution across bill types signals **broad user adoption** with no single utility dominating – indicative of a mature recurring-payment user base. The consistent low failure rate further confirms reliable payment infrastructure for everyday utility transactions.

Bill Categories & Patterns

Electricity Bill

~₹10M · Highest frequency, predictable monthly cadence

DTH

~₹10M · Subscription renewals driving consistent volume

Mobile Bill

~₹10M · Postpaid recharges and plan renewals

Cable TV

~₹10M · Steady despite OTT competition

Failure Pattern

Insufficient Amount accounts for 100% of failed bill payments – mirroring Money Transfer failures and reinforcing the case for platform-wide pre-transaction balance validation.

Monthly baseline: ~₹1.03M/month with a November peak of ₹2.03M. No major seasonal spikes unlike Loans or Insurance.

Key Insights & Recommendations

Platform Diagnostics

96.75%+ Success Rate

Platform reliability is strong. The 3.25% failure rate is concentrated in three fixable root causes: Wrong PIN, Wrong Info, and Server Errors – none structural.

Loans Drive Highest Value

Loan services contribute the highest transaction amounts across all verticals – a key revenue segment to protect and scale with targeted reliability investments.

Seasonal Peaks: July & November

Both months consistently record 2× baseline transaction volume. Infrastructure scaling should be planned ahead of these windows to avoid Server Error spikes.

Actionable Fixes

01

Improve server uptime at peak hours

Server Error causes 25.70% of all failures platform-wide and 33.60% in Insurance – direct uptime investment yields outsized reliability gains.

02

Add pre-transaction balance validation

Insufficient Amount drives 100% of Money Transfer and Bills failures – a single validation layer eliminates this category entirely.

03

Strengthen PIN/info validation UX

Wrong PIN (27.52%) and Wrong Info (27.61%) together account for over 55% of failures – better inline UX prompts reduce these at no infrastructure cost.

04

Optimize banking integrations

Bank Denied failures (5.83%) require deeper integration-layer debugging with partner banks for smoother resolution pathways.

Skills Demonstrated

EXCEL

POWER BI

POWER QUERY

DAX

DATA CLEANING

KPI REPORTING

DASHBOARD DESIGN

BUSINESS ANALYTICS

Sources

Project Report

[PhonePe_Project_Report.pdf](#)